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EDUCATION

Bachelor of Science in Computer Convergence Software — Mar 2019 – Feb 2025

- Korea University Sejong Campus, Korea
- GPA: 4.21/4.5 (Major: 4.36/4.5)
- Relevant Coursework: Computational Linear Algebra, Numerical Analysis, Discrete Mathematics, Probability and Statistics, Algorithms, Operating Systems, Convergence Informatics**,** AI Robotics

PROFESSIONAL EXPERIENCES

RASTECH Robot Laboratory — Research Engineer — Jan 2025 –

- Developed entire task and motion planning and motion control, and perception pipelines for mobile humanoid robots
- Developed asynchronous backend system for real-time data communication and telemetry across heterogeneous robots

Network Management Laboratory, KU Sejong Campus — Undergraduate Researcher — Sep 2023 - Dec 2024

- Supervisor : Dr. Myung-Sup Kim
- Project : Developed GUI Application for Visualizing Global Path Planner of Autonomous Vehicles

Artificial Intelligence System Laboratory, KU Sejong Campus — Undergraduate Researcher — Mar 2022 – Mar 2023

- Supervisor : Dr. Min-Hoe Kim
- Project : Develop a Voiced-Based Indoor Navigation Robot for the Visually impaired

RESEARCH INTERESTS

Main field: Intelligent & Autonomous Robotic Systems

Current focus: VLM (VLA) based Task and Motion Planning (TAMP), Dexterous Object Grasping, Sim2Real Transfer, Domain Adaptation

PUBLICATION

1. Y. Jang, J. Choi, **Y. Kim**, J. Park and M. Kim, "Pre-processing Trends in Public Datasets for Application Traffic Classification," *Proceedings of the 2024 KICS Winter Conference,* pp. 1054-1055, 2024

SKILLS

- Robotics: ROS2, Movelt2, Gazebo, Isaac Sim, Dora, ROS2 Control
- Programming languages: Python, C++
- Containerization : Docker, Docker Compose
- Full-stack & other Frameworks : FastAPI, Flask, PyQt5, Langchain
- English:
  - TOEIC: 885/990,
  - TEPS: 339/600

CERTIFICATIONS & GROUP ACTIVITIES

- AI Robotics KR[AIRK] Group Operating Manager, 2027
- Course Certificate in Advanced Robot Technology and Its Application III, Seoul National University (via K-MOOC), 2026

HONORS AND AWARDS

- **National Science and Technology Scholarship (Full Tuition Scholarship)** 2025
- **1st Place, KT Autonomous Driving Bootcamp Competition** 2023
- Graduation with Honors (Great Honored Student) 2023
- 3rd Place, Department of Computer Convergence Software's Academic Festival (Korea University Sejong Campus) 2019

PROJECTS

Development of a Dual-Arm Humanoid Task Execution System Integrated with Transformer-Based Perception Feb 2026 – Present

- Kinematic Modeling from CAD : Optimized and generated URDF/SRDF models from CAD designs for a 37-DoF mobile humanoid robot, and integrated **synthesized 3D meshes via Google Nano Banana** to make realistic simulation environments in both **Gazebo** and **Isaac Sim**.
- Hardware Interface & Control : Developed custom **ros2_control** hardware interfaces to bridge high-level motion planning with low-level actuator commands, **ensuring stability, real-time by utilizing CAN-FD communication**.
- Object Grasping & Manipulation: Engineered an **end-to-end task execution workflow** integrating **Foundation Pose, RT-DETR, and MoveIt2** to successfully execute robust grasp-and-manipulation episodes for diverse objects.
- Automated Data Process Pipeline: Built an automated image processing and **auto-labeling** pipeline leveraging **OWL-ViT**.

Development of a Teleoperation Framework for Mobile Humanoid Robots Mar 2026 – Sep 2026

- Meta Quest 3 Teleoperation Pipeline**:** Architected **a low-latency, real-time teleoperation** pipeline through Meta Quest 3 **for intuitive 10-DoF hand and 7-DoF arm control,** leveraging the **DORA.rs**

Development of a Bridge for Heterogeneous Robot Management Apr 2025 – Sep 2026

- Asynchronous Backend Server: Built an **asynchronous FastAPI backend** framework to achieve **low-latency telemetry data communication** across multiple robots.
- Unified Multi-Robot Data Integration: Developed a standardized data protocol to consolidate diverse robotic telemetry, building an integrated backend system that enhances interoperability across the entire heterogeneous robot fleet

Development of a Motion System for Character Robot Mar 2025 – Dec 2025

- Kinematic Modeling from CAD: Generated URDF/SRDF configurations from structural CAD files for 6-DoF dual-arm character robots (**Kumdori** and **Kumsuni, Daejeon City Mascots**).
- Control Infrastructure Setup: Integrated ros2_control packages to directly map high-level motion profiles to low-level actuator commands.
- Interactive Motion Generation: **Designed and deployed interactive motion profiles** to execute engaging service behaviors.

Implementation of RAG-based Tool-calling Chat Module for Enhanced Robot Service July 2025 – Dec 2025

- Engineered a API-based tool-calling system to trigger external services (e.g., Weather, Places) and robotic control frameworks including Navigation2, MoveIt2.
- Implemented **Pydantic-based verification** layers **to parse and validate LLM-generated outputs** into reliable, structured robot control commands.

Implementation of a GUI Tool for Global Path Plan of Autonomous Sweeper Vehicles Feb 2025 – Dec 2025

- Developed a Lanelet2 parsing module to convert high-definition map data into structured road network representations for global routing.
- Built a cross-platform, MVC-based PyQt5 GUI integrated with the Kakao Maps API to visualize real-time global paths for multiple autonomous cleaning vehicles.

SUMMARY

I am a robotics engineer specializing in motion planning, real-time control, and system integration for mobile manipulation platforms. My professional expertise spans the entire development pipeline—from kinematic modeling to deploying modular, hardware-agnostic control frameworks that ensure scalability across heterogeneous robotic platforms.

Building on this foundation, my research vision focuses on **bridging the gap** between **advanced AI technologies** and **physical robotics** to solve **complex, unstructured tasks** in **real-world environments**. To make this possible, I intend to dedicate my studies to leveraging **Foundation Models (VLMs/VLAs), Data-efficient Learning** within high-fidelity **simulations environments**.

Ultimately, I aspire to develop robust, scalable robotic intelligence that overcomes the limitations of traditional control methods and delivers direct technological value to actual industries.

REFERENCES

- Dr. Myung-Sup, Kim, Korea University Sejong Campus
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